Design and Performance Analysis of an Adaptive RMS-Based Voltage Disturbance Detection System

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***Abstract*—Power quality disturbances such as voltage sag, swell, and interruption can affect the reliable operation of electrical and electronic equipment. Accurate and timely detection of these disturbances is therefore important for power system monitoring. This project presents the design and performance analysis of an adaptive Root Mean Square (RMS)-based voltage disturbance detection system using MATLAB. The proposed method employs a multi-level window selection strategy to balance estimation accuracy and detection speed. A 50-sample least-squares estimator is used to identify changes in the fundamental voltage magnitude, while 100-sample and 200-sample windows are used for fast and steady-state RMS estimation, respectively. The estimated RMS voltage is converted into per-unit values and classified using predefined thresholds to identify normal operation, sag, swell, and interruption conditions. The system is evaluated using simulated voltage waveforms under clean conditions, different Gaussian noise levels, and harmonic distortion containing third, fifth, and seventh harmonics. Performance is assessed using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and classification accuracy. Simulation results indicate that the adaptive approach provides lower estimation errors and higher classification accuracy than the fixed-window RMS method under the tested conditions. The results demonstrate the potential of multi-level adaptive RMS estimation for efficient voltage disturbance monitoring.**

***Index Terms*—Adaptive RMS Estimation, Power Quality Monitoring, Voltage Sag, Voltage Swell, Voltage Interruption, Multi-Level Window Selection, Digital Signal Processing, Harmonic Distortion**

I. Introduction

Electrical power quality plays an important role in the reliable operation of modern electrical and electronic equipment. In practical power systems, the supply voltage may experience variations due to sudden load changes, switching operations, faults, and other disturbances. Such variations can affect the performance of sensitive equipment and may lead to malfunction or interruption of industrial and domestic applications.

Among the commonly encountered power quality disturbances, voltage sag, voltage swell, and interruption are particularly significant. Voltage sag refers to a temporary reduction in the RMS voltage magnitude, whereas voltage swell represents a temporary increase. Voltage interruption occurs when the supply voltage falls to a very low level. The accurate detection of these disturbances is therefore necessary for effective power quality monitoring and analysis [3], [5], [9].

The Root Mean Square (RMS) value is widely used to represent the effective magnitude of an alternating voltage signal. A conventional fixed-window RMS method calculates the voltage magnitude over a predefined number of samples. Although this approach provides a simple estimation technique, the use of a fixed window introduces a trade-off between estimation accuracy and response speed. A longer window provides more stable RMS estimates but may delay the identification of sudden voltage changes. A shorter window can respond faster but may produce greater estimation variations.

To address this limitation, this work presents an adaptive RMS-based voltage disturbance detection system implemented in MATLAB. The proposed approach uses a multi-level window selection strategy consisting of a 50-sample least-squares estimator for detecting changes in the fundamental voltage magnitude, a 100-sample window for faster RMS estimation, and a 200-sample window for steady-state estimation. Based on the detected changes, the estimation window is selected to improve the response of the system during voltage disturbances while retaining stable estimation during normal operating conditions.

The proposed method is evaluated through simulated voltage waveforms containing sag, swell, and interruption conditions. Additional experiments are conducted under different Gaussian noise levels and harmonic distortion involving third, fifth, and seventh harmonic components. The performance of the adaptive method is compared with that of a conventional fixed-window RMS estimator using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and classification accuracy.

The objective of this study is to investigate whether adaptive window selection can improve RMS estimation and voltage disturbance classification compared with a fixed-window approach under the simulated operating conditions. Previous studies have examined voltage sag monitoring, detection, and classification methods [1], [2], [5], [7], [9].

II. Problem Statement and Objectives

A. Problem Statement

Conventional RMS-based voltage monitoring systems commonly employ a fixed-size sampling window for estimating the effective voltage magnitude. Although this approach is simple to implement, it introduces a limitation in detecting sudden voltage variations. A larger estimation window provides stable

RMS values but requires more time to reflect changes in the input signal. In contrast, a smaller window improves the response to voltage variations but may produce less stable estimates.

This trade-off can affect the timely identification of power quality disturbances such as voltage sag, swell, and interruption. Therefore, an estimation technique that can adapt its window length according to changes in the input voltage is desirable.

The problem addressed in this work is to develop and evaluate an adaptive RMS estimation technique that improves disturbance detection response while maintaining estimation accuracy under different simulated signal conditions.

B. Objectives

The primary objectives of this project are:

- To develop a MATLAB-based voltage disturbance detection system for identifying voltage sag, swell, and interruption.
- To implement a conventional fixed-window RMS estimator using a 200-sample window.
- To design an adaptive RMS estimation approach using 100-sample and 200-sample windows.
- To employ a 50-sample least-squares estimator for detecting changes in the fundamental voltage magnitude.
- To classify voltage conditions using predefined per-unit voltage thresholds.
- To compare fixed and adaptive RMS estimation using RMSE, MAE, and classification accuracy.
- To evaluate the performance of both methods under Gaussian noise and harmonic distortion.

III. Theoretical Background

A. Root Mean Square Estimation

The Root Mean Square (RMS) value represents the effective magnitude of an alternating voltage signal. For a discrete-time voltage signal containing N samples, the RMS value is calculated as

$$V_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} v^2[n]} \tag{1}$$

where $v[n]$ represents the sampled voltage signal and N denotes the number of samples considered for estimation.

In this work, the conventional fixed-window method uses 200 samples, corresponding to one complete cycle of a 50 Hz supply signal sampled at 10 kHz. The adaptive method uses a shorter 100-sample window during detected voltage changes and the 200-sample window during steady-state operation.

B. Voltage Disturbances

Voltage disturbances are temporary variations in the RMS magnitude of the supply voltage. This work considers three major disturbances.

1) Voltage Sag: Voltage sag is a temporary reduction in the RMS voltage magnitude. In this project, a sag condition is simulated by reducing the nominal voltage magnitude to 50% for a specified time interval.

2) Voltage Swell: Voltage swell is a temporary increase in the RMS voltage magnitude. The simulated swell condition increases the nominal voltage magnitude to 130%.

3) Voltage Interruption: Voltage interruption occurs when the supply voltage decreases to a very low magnitude. In the simulation, the voltage is reduced to 2% of its nominal value to represent an interruption condition.

C. Per-Unit Voltage Representation

The per-unit (pu) system expresses the estimated RMS voltage relative to its nominal value. It is calculated as

$$V_{\mathrm{pu}} = \frac{V_{\mathrm{RMS}}}{V_{\mathrm{nominal}}} \tag{2}$$

where V_{RMS} is the estimated RMS voltage and V_{nominal} is the nominal RMS voltage.

The per-unit representation allows voltage disturbances to be classified using predefined thresholds. The classification limits used in this project are selected simulation thresholds rather than universal limits, and are given in Table I. Power-quality monitoring practices and disturbance terminology are discussed in relevant standards [3].

TABLE I
Voltage Disturbance Classification Thresholds

Voltage Range (pu)	Condition
$V_{\mathrm{pu}} < 0.1$	Interruption
$0.1 \le V_{\mathrm{pu}} < 0.9$	Sag
$0.9 \le V_{\mathrm{pu}} \le 1.1$	Normal
$V_{\mathrm{pu}} > 1.1$	Swell

D. Least-Squares Fundamental Estimation

A least-squares estimator is used to estimate the fundamental voltage magnitude over a short observation window. The sampled voltage is represented using sine and cosine components:

$$v[n] = A\sin(\omega t_n) + B\cos(\omega t_n) \tag{3}$$

where A and B are the estimated coefficients and ω is the angular supply frequency.

The coefficients are obtained by minimizing the squared error between the sampled voltage and the model. Least-squares fundamental estimation is used here as a short-window change indicator; it is distinct from the total RMS calculation used for final voltage estimation. The corresponding fundamental RMS magnitude is calculated as

$$V_{\mathrm{fund,RMS}} = \frac{\sqrt{A^2 + B^2}}{\sqrt{2}} \tag{4}$$

In the proposed system, this estimator operates on a 50-sample window and is used for detecting changes in the fundamental voltage magnitude. It is not used as the final total RMS output.

IV. Proposed Methodology

A. System Overview

The proposed system detects voltage sag, swell, and interruption using an adaptive RMS estimation technique. The system is implemented in MATLAB and operates on a simulated single-phase voltage signal with a nominal frequency of 50 Hz and an RMS voltage of 230 V.

The methodology consists of voltage signal generation, fixed-window RMS estimation, short-window fundamental magnitude estimation, adaptive window selection, per-unit conversion, and disturbance classification. The performance of the proposed method is evaluated by comparing its output with that of a conventional fixed-window RMS estimator.

B. Fixed-Window RMS Estimation

The conventional method uses a fixed observation window of 200 samples to calculate the RMS voltage. At a sampling frequency of 10 kHz and a supply frequency of 50 Hz, this window corresponds to one complete fundamental cycle.

The fixed-window RMS estimate is calculated using

$$V_{\text{fixed}}[n] = \sqrt{\frac{1}{200}\sum_{k=0}^{199} v^2[n-k]} \tag{5}$$

where $v[n]$ represents the sampled input voltage.

Although the fixed-window estimator provides a stable RMS estimate, its response to sudden voltage variations is limited by the observation window length. Sliding-window RMS approaches and fast voltage-sag detection methods have also been investigated in the literature [7], [13].

C. Short-Window RMS Estimation

To obtain a faster response during voltage variations, a shorter window of 100 samples is employed. The corresponding RMS estimate is calculated as

$$V_{\text{fast}}[n] = \sqrt{\frac{1}{100}\sum_{k=0}^{99} v^2[n-k]} \tag{6}$$

This window represents half a cycle of the 50 Hz input signal. It provides a faster response than the 200-sample estimator, although its estimate may exhibit greater variation.

D. Least-Squares Change Detection

A 50-sample least-squares estimator is used to estimate the fundamental component of the input voltage. The sampled signal is represented using sine and cosine components, and the coefficients are obtained by solving a least-squares problem.

The fundamental RMS magnitude is calculated using

$$V_{\text{det}}[n] = \frac{\sqrt{A^2[n] + B^2[n]}}{\sqrt{2}} \tag{7}$$

where $A[n]$ and $B[n]$ are the estimated sine and cosine coefficients.

The difference between the short-window fundamental estimate and the fixed-window RMS estimate is monitored:

$$D[n] = |V_{\text{det}}[n] - V_{\text{fixed}}[n]| \tag{8}$$

A change is identified when this difference exceeds a predefined threshold of 5% of the nominal RMS voltage. For a nominal voltage of 230 V, the threshold is 11.5 V.

E. Adaptive Window Selection

The adaptive estimation method selects between the 100-sample and 200-sample RMS windows based on the detected voltage changes.

When the change detection condition is satisfied, the system selects the 100-sample RMS estimate for a duration of 100 samples. This provides a faster response during the detected variation. After this interval, the system returns to the 200-sample RMS estimate for steady-state operation.

A refractory interval of 50 samples is also employed to prevent repeated triggering within a short interval.

The adaptive RMS output can be represented as

$$V_{\text{adaptive}}[n] = \begin{cases} V_{\text{fast}}[n], & n \le n_{\text{fast}} \\ V_{\text{fixed}}[n], & \text{otherwise} \end{cases} \tag{9}$$

where n_{fast} represents the last sample of the fast estimation interval.

F. Disturbance Classification

After RMS estimation, the voltage magnitude is converted into per-unit form using the nominal RMS voltage of 230 V.

The resulting per-unit voltage is classified into four operating conditions: normal, sag, swell, and interruption. The thresholds used for classification are presented in Section III.

The classification process is independently applied to the fixed-window and adaptive RMS outputs to enable a direct comparison of their detection performance.

G. Harmonic Distortion Evaluation

The proposed estimation approach is additionally evaluated using voltage signals containing third, fifth, and seventh harmonic components. Harmonic measurement and control are addressed in power-quality standards [4]. The harmonic amplitudes are set to 15%, 10%, and 5% of the fundamental component, respectively.

For this experiment, the fundamental magnitude estimates separated by one complete 200-sample cycle are compared to identify changes in the fundamental voltage magnitude. This modification helps reduce false triggering caused by periodic variations in the short-window harmonic estimates.

The total RMS reference includes the fundamental and harmonic components. The fixed and adaptive estimation outputs are then evaluated against this reference. Recent work has also explored signal-processing and learning-based approaches for power-quality disturbance detection and classification [6], [8], [10]–[12], [14], [15].

V. MATLAB Implementation

A. Simulation Parameters

The proposed adaptive RMS-based voltage disturbance detection system is implemented and evaluated using MATLAB. A single-phase sinusoidal voltage signal is generated with a fundamental frequency of 50 Hz and a nominal RMS voltage of 230 V. The sampling frequency is selected as 10 kHz, and the total simulation duration is 1 second.

The simulation parameters used in the implementation are summarized in Table II.

TABLE II
MATLAB Simulation Parameters

Parameter	Value
Supply frequency	50 Hz
Sampling frequency	10 kHz
Simulation duration	1 second
Nominal RMS voltage	230 V
Fixed RMS window	200 samples
Fast RMS window	100 samples
Change detection window	50 samples
Change detection threshold	5% of nominal RMS
Hold duration	100 samples
Refractory interval	50 samples

B. Voltage Signal Generation

A sinusoidal voltage waveform is generated using the nominal peak voltage and fundamental angular frequency. The peak voltage is calculated from the nominal RMS voltage as

$$V_{\text{peak}} = \sqrt{2} V_{\text{nominal}} \tag{10}$$

The input voltage is initially maintained at its nominal value. Three disturbances are introduced at different time intervals to evaluate the detection performance.

The voltage sag is introduced from 0.20 s to 0.35 s by reducing the voltage magnitude to 50% of its nominal value. A voltage swell is introduced from 0.50 s to 0.65 s by increasing the magnitude to 130%. Finally, an interruption condition is simulated from 0.75 s to 0.85 s by reducing the voltage magnitude to 2% of its nominal value.

These conditions provide a controlled test environment for evaluating the fixed and adaptive RMS estimation methods.

C. Noise and Harmonic Test Conditions

To investigate noise robustness, Gaussian noise is added to the simulated voltage signal at different signal-to-noise ratio (SNR) levels. The noise-free condition and SNR levels of 30 dB, 20 dB, and 10 dB are considered.

Harmonic distortion is evaluated by adding third, fifth, and seventh harmonic components to the fundamental voltage waveform. Their amplitudes are selected as 15%, 10%, and 5% of the fundamental component, respectively.

The noise and harmonic experiments are conducted separately to examine the behaviour of the proposed method under different signal conditions.

D. Performance Evaluation Metrics

The performance of the fixed and adaptive RMS estimators is evaluated using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). Classification accuracy is also calculated to assess the ability of each method to identify the correct voltage condition.

The RMSE is calculated as

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(V_i - \hat{V}_i)^2} \tag{11}$$

The MAE is calculated as

$$\text{MAE} = \frac{1}{M}\sum_{i=1}^{M}|V_i - \hat{V}_i| \tag{12}$$

where V_i represents the ideal RMS reference, $\hat{V}_i$ represents the estimated RMS value, and M denotes the number of valid samples considered for evaluation. The reference RMS waveform is obtained from the known voltage-magnitude profile used to generate the simulated signal and serves as the ground-truth value for comparison.

Classification accuracy is calculated as

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}} \times 100 \tag{13}$$

where N_{correct} is the number of correctly classified samples and N_{total} is the total number of valid samples.

VI. Results and Discussion

A. Performance under Clean Conditions

The first experiment evaluates the performance of the fixed and adaptive RMS estimation methods under clean voltage conditions. The input waveform contains voltage sag, swell, and interruption events at the specified time intervals.

The performance is evaluated using RMSE, MAE, and classification accuracy. The results obtained from the MATLAB simulation are presented in Table III.

TABLE III
Performance under Clean Conditions

Metric	Fixed RMS	Adaptive RMS
RMSE (V)	31.4354	23.3276
MAE (V)	8.3472	4.1723
Accuracy (%)	92.44	96.38

The adaptive RMS method provides lower RMSE and MAE values than the conventional fixed-window approach. The classification accuracy also improves from approximately 92.44% to 96.38%. These results indicate that adaptive window selection improves the estimation and classification performance for the simulated clean voltage conditions. The RMSE is higher than the MAE because squaring the error gives greater weight to large deviations, particularly those occurring around disturbance transitions.

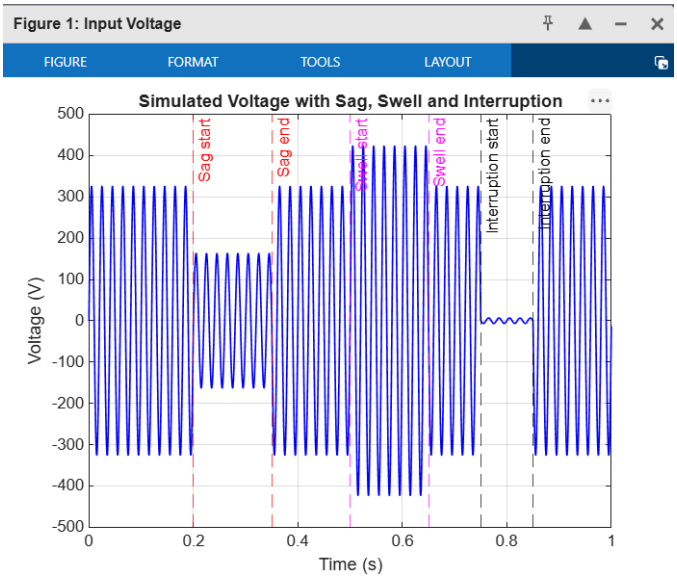


Fig. 1. Simulated Voltage Waveform with Sag, Swell and Interruption

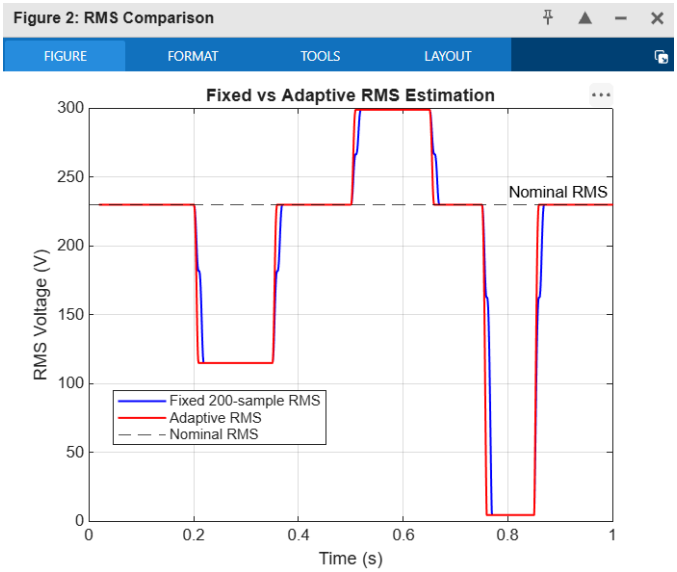


Fig. 2. Comparison of Fixed and Adaptive RMS Estimation

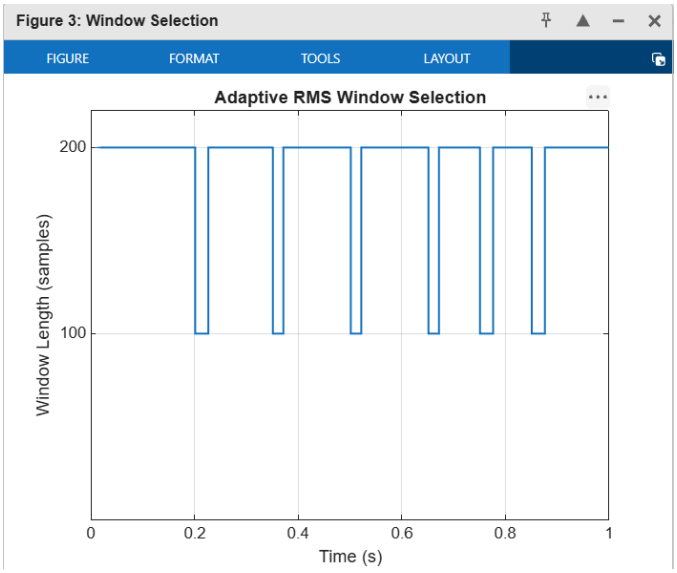


Fig. 3. Adaptive RMS Window Selection

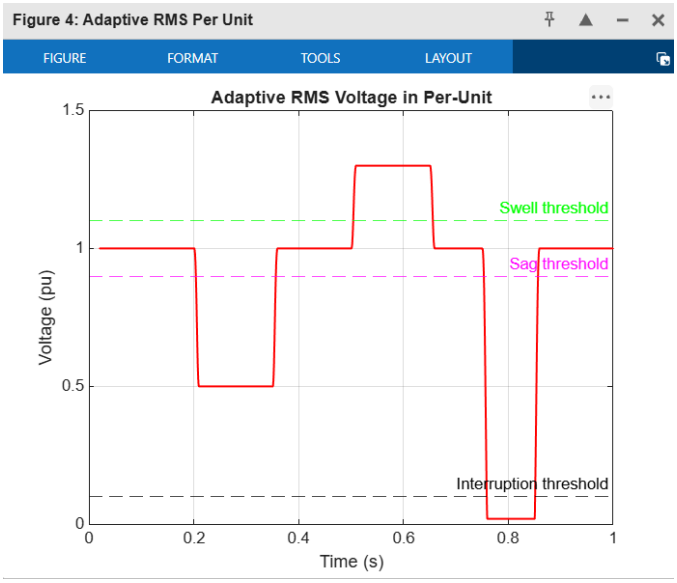


Fig. 4. Adaptive RMS Voltage in Per-Unit

The simulated input waveform and the fixed-versus-adaptive RMS estimates are shown in Figs. 1 and 2. The selected window lengths, per-unit voltage profile, disturbance classification, and estimation errors are illustrated in Figs. 3–6.

B. Noise Robustness Analysis

The second experiment investigates the behaviour of both estimation methods under different Gaussian noise levels. The noise-free condition and SNR values of 30 dB, 20 dB, and 10 dB are considered.

The RMSE, MAE, and classification accuracy obtained at each noise level are summarized in Table IV.

The results show that the adaptive method maintains lower estimation errors and higher classification accuracy than the fixed-window method at all the tested noise levels. As the SNR decreases, the estimation errors increase and classification accuracy decreases for both approaches. Nevertheless, the adaptive method maintains better performance under the simulated noise conditions.

The variation in estimation error and classification accuracy with SNR is shown in Figs. 7 and 8, respectively.

C. Harmonic Distortion Analysis

The third experiment evaluates the estimation methods in the presence of harmonic distortion. The input voltage signal includes third, fifth, and seventh harmonic components with amplitudes of 15%, 10%, and 5% of the fundamental component, respectively.

TABLE IV
PERFORMANCE COMPARISON UNDER DIFFERENT NOISE LEVELS

SNR (dB)	RMSE (V)		MAE (V)		Accuracy (%)	
	Fixed	Adaptive	Fixed	Adaptive	Fixed	Adaptive
Clean	31.4354	23.3276	8.3472	4.1723	92.44	96.38
30	31.4307	23.3380	8.9556	4.8658	92.45	96.40
20	31.7081	23.7555	10.8859	7.0119	91.13	94.39
10	37.5396	31.9360	22.4725	19.7479	84.51	87.08

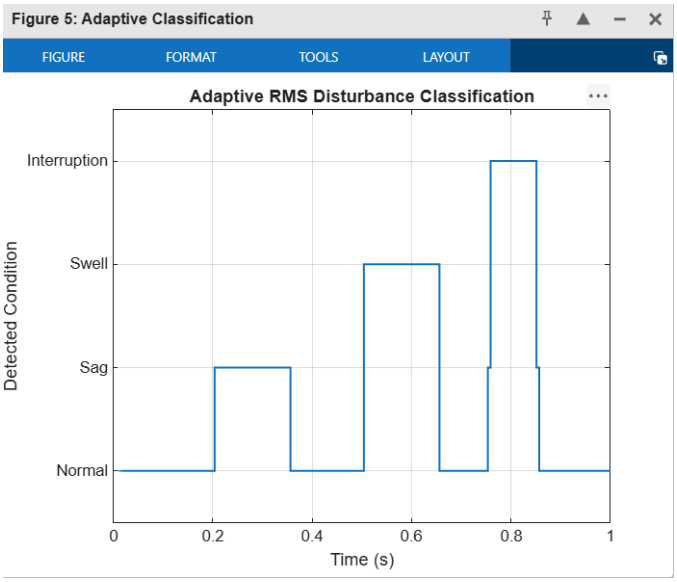


Fig. 5. Adaptive Voltage Disturbance Classification

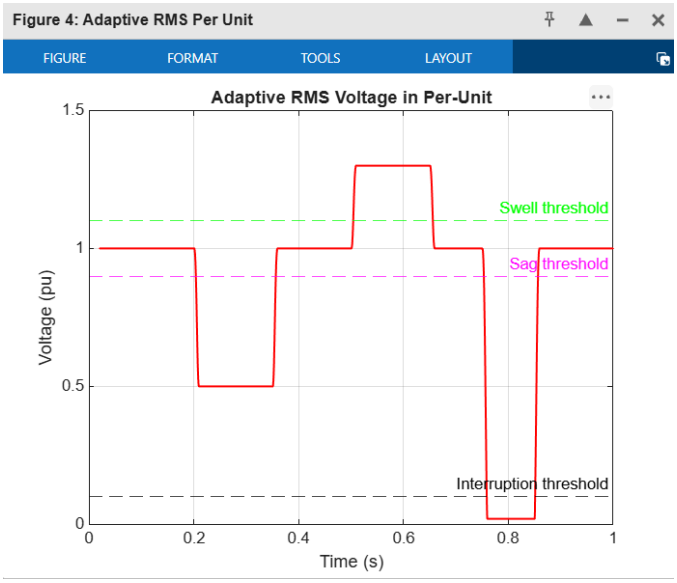


Fig. 6. Comparison of Fixed and Adaptive RMS Estimation Errors

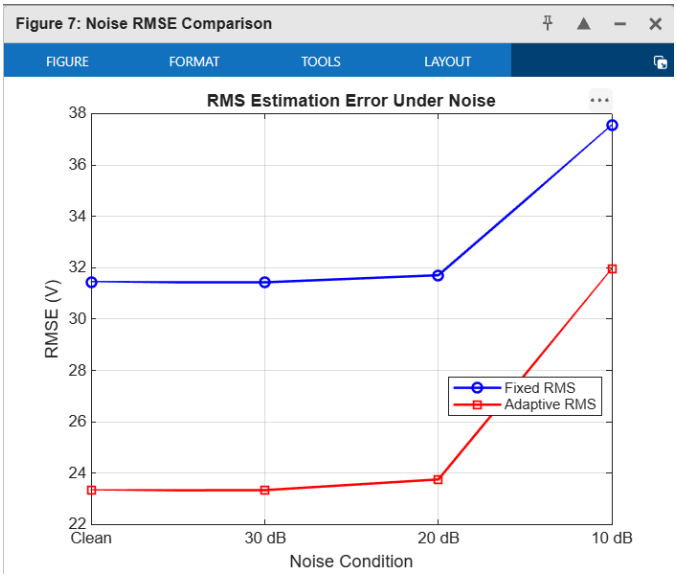


Fig. 7. RMS Estimation Error under Different Noise Levels

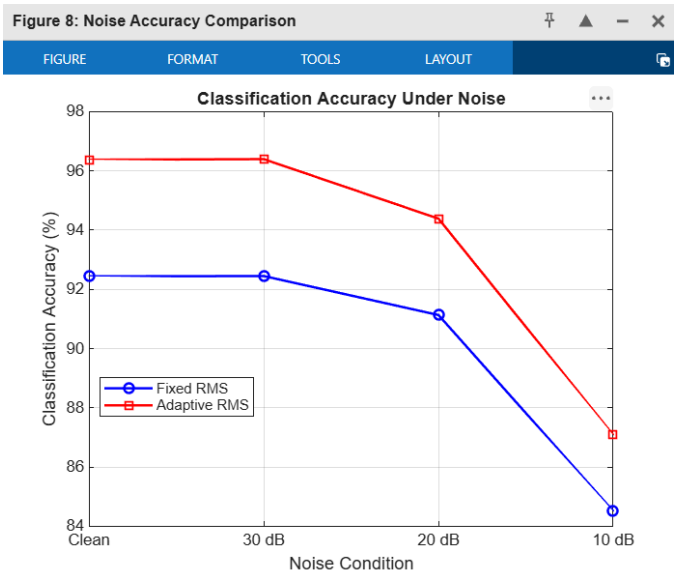


Fig. 8. Classification Accuracy under Different Noise Levels

Since the harmonic components contribute to the total RMS magnitude, the reference RMS value is adjusted to account for their contribution. The resulting nominal RMS reference is approximately 233.99 V.

The performance obtained under harmonic distortion is presented in Table V.

The adaptive method achieves lower RMSE and MAE values compared with the fixed-window estimator. Its classification accuracy is also higher under the tested harmonic

TABLE V
PERFORMANCE UNDER HARMONIC DISTORTION

Metric	Fixed RMS	Adaptive RMS
RMSE (V)	31.6436	23.0656
MAE (V)	8.4920	4.2441
Accuracy (%)	92.37	96.30

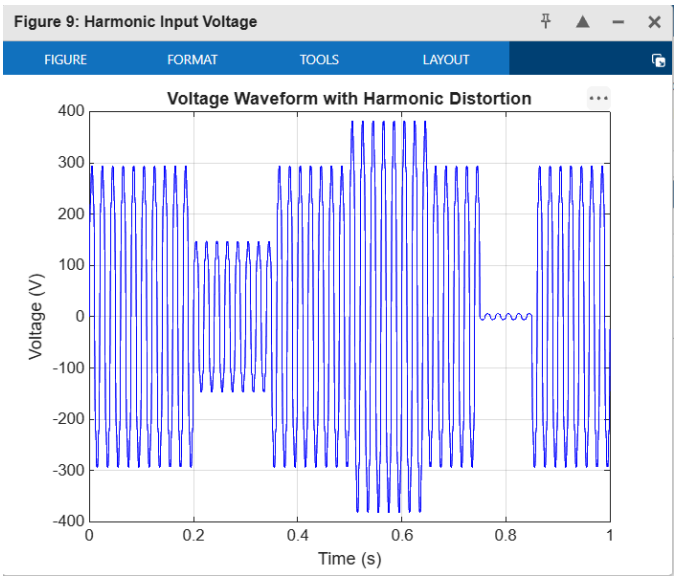


Fig. 9. Voltage Waveform with Harmonic Distortion

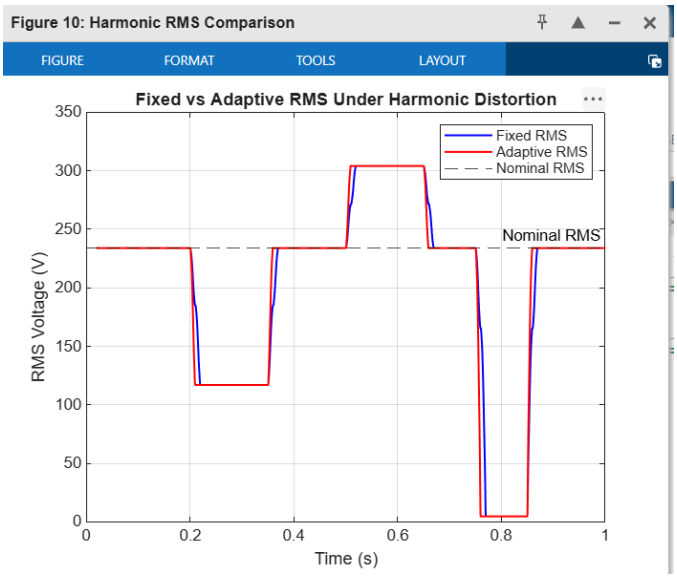


Fig. 10. Fixed and Adaptive RMS Estimation under Harmonic Distortion

conditions. The results suggest that the modified change detection approach helps reduce false triggering caused by periodic variations in short-window harmonic estimates.

The distorted input waveform, RMS estimates, per-unit voltage, and selected window lengths are presented in Figs. 9–12.

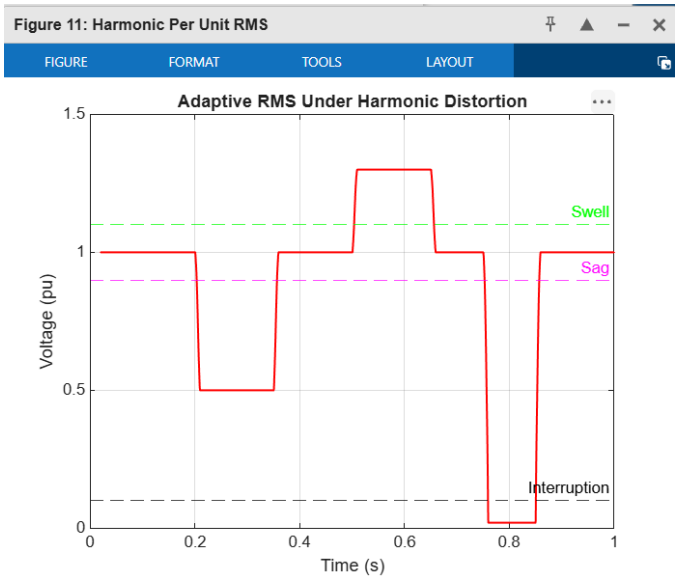


Fig. 11. Adaptive RMS Voltage under Harmonic Distortion

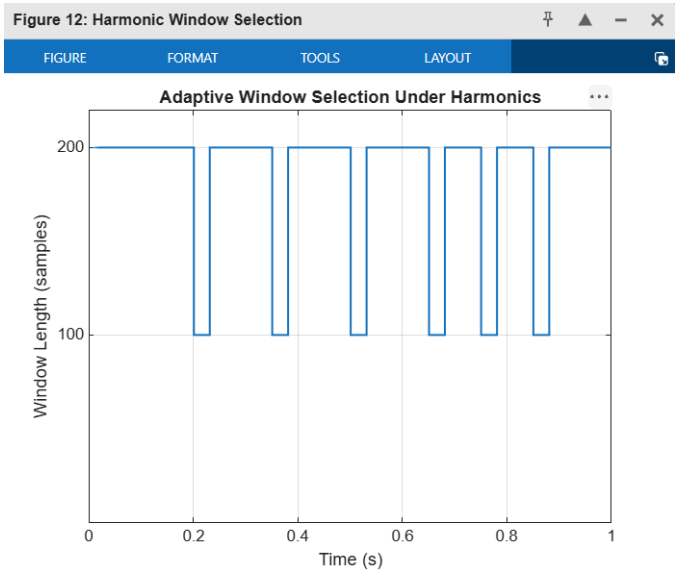


Fig. 12. Adaptive Window Selection under Harmonic Distortion

D. Comparative Performance Analysis

Across the three experiments, the adaptive RMS method provides lower estimation errors and higher classification accuracy than the fixed-window approach.

The improvement is observed under clean conditions, different Gaussian noise levels, and harmonic distortion. The shorter estimation window enables a faster response during detected voltage changes, while the longer window is retained during steady-state operation.

However, the results are based on simulated voltage signals and predefined disturbance conditions. Further evaluation using practical power-system measurements is required to establish the performance of the method under real operating

conditions.

VII. Conclusion and Future Scope

This work presented the design and performance analysis of an adaptive RMS-based voltage disturbance detection system using MATLAB. The proposed approach employs a multi-level window selection strategy to improve the estimation and classification of voltage variations. A 50-sample least-squares estimator is used for detecting changes in the fundamental voltage magnitude, while 100-sample and 200-sample windows are used for fast and steady-state RMS estimation, respectively.

The system was evaluated under clean voltage conditions, Gaussian noise, and harmonic distortion involving third, fifth, and seventh harmonic components. The simulation results indicate that the adaptive method provides better performance than the conventional fixed-window RMS estimator under the tested conditions. Under clean conditions, the adaptive method achieved an RMSE of 23.3276 V and an MAE of 4.1723 V, compared with 31.4354 V and 8.3472 V, respectively, for the fixed-window method. Classification accuracy also improved from 92.44% to 96.38%.

Similar improvements were observed in the noise and harmonic distortion experiments. These results demonstrate the potential of adaptive window selection for improving voltage disturbance monitoring in the simulated environment.

The present work is limited to MATLAB-based simulation. Future work can focus on validating the proposed method using real power-system voltage measurements, investigating its performance under different disturbance types, and implementing the algorithm on real-time hardware. Further improvements may also include advanced signal processing techniques for more reliable disturbance detection under complex operating conditions.

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